

The $\mathfrak{A}^{(1)\dagger} \bar{\mathfrak{C}}^\dagger \times \mathfrak{B}_2$ interference

Full check, the divergence structure, and why the $+$ prescription fails here

$$S = p^+ + k^+, \quad A = (x - z)^2, \quad B = (x - w)^2, \quad C = (z - w)^2, \quad D = p^+ A + k^+ B, \quad \mathcal{V}_w = \frac{z-w}{C} - \frac{x-w}{2B}, \quad \mathcal{V}_z = \frac{z-w}{C} + \frac{x-z}{2A}$$

The calculation is right except for one typo (§3), and there is a subtlety in the prefactor that is *also* right but must be stated or it looks wrong (§1). The w' integration, the phase split, and all six longitudinal weights are exact (§4). **On divergences: no UV, and — unlike the two- $\mathfrak{C}$ row — no collinear pole either,** because here the z integral carries the phase $e^{+i \frac{p^+}{k^+} k \cdot z}$, which cuts the would-be collinear logarithm at $|z| \sim k^+ / (p^+ |k_\perp|)$ (§8). So there is nothing to put in $\overline{\text{MS}}$ and no fragmentation subtraction to make here. **But that cutoff grows as $p^+ \rightarrow 0$,** and the result is a **double logarithm**:

$$I(p^+) p^+ = 0.1085 - 0.0490 \log \frac{k^+}{p^+} \quad (\text{stable under a factor 4 in cutoff and grid})$$

$$\implies \int_\Lambda \frac{dp^+}{p^+} \log \frac{k^+}{p^+} \sim \frac{1}{2} \log^2 \frac{k^+}{\Lambda}.$$

Therefore you cannot use the same $+$ prescription here (§10): it needs a finite endpoint value $g(0)$, and here g diverges logarithmically at the endpoint.

1. The prefactor — and the factor hidden in your rewritten $\delta^{(2)}$

Take the three coefficients, all evaluated at the arguments the row uses ($\mathfrak{A}^{(1)}$ and $\mathfrak{C}_1$ at $k_{\text{arg}}^+ = S$):

$$\mathfrak{A}^{(1)\dagger} = -\frac{ig}{\sqrt{2}\pi\sqrt{S}}, \quad \bar{\mathfrak{C}}^\dagger = -\frac{g}{2\pi\sqrt{2S}} \frac{\sqrt{p^+k^+}}{S}, \quad \mathfrak{B}_2 = \frac{ig^2\sqrt{p^+k^+}}{4\pi^2 D S}$$

(the last uses $\sqrt{p^+(S-p^+)} = \sqrt{p^+k^+}$). Multiplying and attaching $\frac{4}{(2\pi)^3}$:

$$\frac{4}{(2\pi)^3} \mathfrak{A}^\dagger \bar{\mathfrak{C}}^\dagger \mathfrak{B}_2 = -\frac{g^4}{32\pi^7} \frac{p^+k^+}{S^3 D} = -\frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{p^+k^+}{S^3 D}$$

while you wrote $-\frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{1}{k^+} \frac{p^+}{S D}$. The ratio is

$$\frac{\text{yours}}{\text{derived}} = \frac{S^2}{k^{+2}} \neq 1 \text{ — but this is not an error.}$$

Where it goes. The definition's delta is $\delta^{(2)}(y' - w' + \frac{p^+}{S}(w' - z))$. You wrote $\delta^{(2)}(\frac{p^+}{k^+}(y' - z) + y' - w')$. These arguments are proportional:

$$\frac{p^+}{k^+}(y' - z) + y' - w' = \frac{S}{k^+} \left[y' - w' + \frac{p^+}{S}(w' - z) \right]$$

(check: multiply out the right side, $\frac{S}{k^+}y' - \frac{S}{k^+}w' + \frac{p^+}{k^+}w' - \frac{p^+}{k^+}z = y' + \frac{p^+}{k^+}y' - w' - \frac{p^+}{k^+}z$ ✓). Since $\delta^{(2)}(\lambda v) = \lambda^{-2}\delta^{(2)}(v)$ in two dimensions, **your delta carries an extra k^{+2}/S^2** — which is exactly cancelled by the S^2/k^{+2} in your prefactor. Net ratio = 1, verified symbolically. Worth a footnote in your draft, or a reader will stop here.

2. The overall minus sign

$\mathfrak{A}^\dagger$ carries $-i$, $\bar{\mathfrak{C}}^\dagger$ carries -1 (it is real), $\mathfrak{B}_2$ carries $+i$. Product: $(-i)(-1)(+i) = i \cdot i = -1$. Your explicit minus is right, and the row is real, as an interference of this type must be once it is added to its conjugate.

3. The tensor contraction — and the one typo

3.1 Setting it up

Write the $\mathfrak{B}_2$ bracket as W^{ji} with the factorised form established earlier:

$$W^{ji} = \delta^{ji} \frac{A-B}{2C} + \frac{S}{k^+} (x-z)^j \mathcal{V}_w^i + \frac{S}{p^+} \mathcal{V}_z^j (x-w)^i$$

The $\mathfrak{C}$ bracket is called as $\bar{\mathfrak{C}}_{1j i k'}$, i.e. with the second and third index slots holding i and k' , whereas the definition $\mathfrak{C}_{1j k i}$ holds k and i . So relabel the definition's $k \rightarrow i$ and $i \rightarrow k'$:

$$[\delta_{im}\delta_{jk} - \frac{S}{p^+}\delta_{jm}\delta_{ik} - \frac{S}{k^+}\delta_{km}\delta_{ij}] \longrightarrow [\delta_{k'm}\delta_{ij} - \frac{S}{p^+}\delta_{jm}\delta_{ik'} - \frac{S}{k^+}\delta_{im}\delta_{jk'}]$$

3.2 Contracting over i, j

Take the three deltas one at a time, exactly as in the previous note:

- $\delta_{k'm}\delta_{ij}W^{ji} = \delta_{k'm}W^{ii} = \delta_{k'm}\text{tr} W$;
- $-\frac{S}{p^+}\delta_{jm}\delta_{ik'}W^{ji} = -\frac{S}{p^+}W^{mk'}$ (set $j = m, i = k'$);
- $-\frac{S}{k^+}\delta_{im}\delta_{jk'}W^{ji} = -\frac{S}{k^+}W^{k'm}$ (set $i = m, j = k'$).

$$R^{mk'} = \delta_{k'm}\text{tr} W - \frac{S}{p^+}W^{mk'} - \frac{S}{k^+}W^{k'm}$$

Note W is **not** symmetric (unlike $T^{mm'}$), so the last two do *not* merge.

3.3 Reading off your six groups

$\text{tr} W = d_{\perp} \frac{A-B}{2C} + \frac{S}{k^+}(x-z) \cdot \mathcal{V}_w + \frac{S}{p^+}\mathcal{V}_z \cdot (x-w)$, and the δ^{ji} part of W contributes $-\frac{S}{p^+}\delta^{mk'} \frac{A-B}{2C} - \frac{S}{k^+}\delta^{k'm} \frac{A-B}{2C}$ from the last two terms. Collecting all the $\delta_{k'm}$ pieces:

$$\delta_{k'm} \left\{ d_{\perp} \frac{A-B}{2C} + \frac{S}{k^+} \left[(x-z) \cdot \mathcal{V}_w - \frac{A-B}{2C} \right] + \frac{S}{p^+} \left[\mathcal{V}_z \cdot (x-w) - \frac{A-B}{2C} \right] \right\}$$

which is **exactly** your groups 1–3, including the extra $-\frac{A-B}{2C}$ inside the second and third brackets that is not in $\mathfrak{B}_2$ itself. Good — that is the tell that you did the contraction and not just a transcription.

The non- δ pieces:

$$\begin{aligned} -\frac{S}{p^+}W^{mk'} \Big|_{\text{non-}\delta} &= -\frac{S^2}{p^+k^+}(x-z)^m \mathcal{V}_w^{k'} - \frac{S^2}{p^{+2}} \mathcal{V}_z^m (x-w)^{k'} \\ -\frac{S}{k^+}W^{k'm} \Big|_{\text{non-}\delta} &= -\frac{S^2}{k^{+2}}(x-z)^{k'} \mathcal{V}_w^m - \frac{S^2}{p^+k^+} \mathcal{V}_z^{k'} (x-w)^m \end{aligned}$$

group	should be	you wrote
$-\frac{S^2}{p^{+2}}$	$\mathcal{V}_z^m (x-w)^{k'} = \frac{(z-w)^m (x-w)^{k'}}{C} + \frac{(x-z)^m (x-w)^{k'}}{2A}$	$\frac{(x-w)^{k'} (z-w)^m}{C} + \frac{(x-z)^m (z-w)^{k'}}{2A}$
$-\frac{S^2}{k^{+2}}$	$(x-z)^{k'} \mathcal{V}_w^m$	same ✓
$-\frac{S^2}{p^+k^+}$	$(x-z)^m \mathcal{V}_w^{k'} + \mathcal{V}_z^{k'} (x-w)^m$, all four terms	same ✓

The one error. In the $-\frac{S^2}{p^{+2}}$ group your second term reads

$$\frac{(x-z)^m (z-w)^{k'}}{2(x-z)^2} \quad \text{but it must be} \quad \frac{(x-z)^m (x-w)^{k'}}{2(x-z)^2}.$$

The reason is structural: that whole group is $-\frac{S^2}{p^{+2}} \mathcal{V}_z^m (x-w)^{k'}$, a single outer product — so **both** of its terms must carry the *same* second factor $(x-w)^{k'}$; only the first factor, $\mathcal{V}_z^m$, splits into two pieces. Numerically, with your version $\|R_{\text{yours}} - R_{\text{exact}}\| = 6.51, 1.98, 0.77$ on three random configurations; with the correction, 10^{-14} . Groups 5 and 6 are right, so this is an isolated slip.

4. The w' integration

Because you rescaled the delta, its w' -coefficient is -1 , so the **Jacobian is exactly 1** (no $(S/k^+)^2$ here — that factor is already sitting in your prefactor, §1). Solving,

$$w' = y' + \frac{p^+}{k^+}(y' - z) \implies w' - z = \frac{S}{k^+}(y' - z), \quad w' - w = (y' - w) + \frac{p^+}{k^+}(y' - z)$$

so the kernel rescales, $\frac{(w'-z)^m}{(w'-z)^2} = \frac{k^+}{S} \frac{(y'-z)^m}{(y'-z)^2}$, and the phase splits exactly as you wrote:

$$e^{-ik\cdot(w'-w)} = e^{-ik\cdot(y'-w)} e^{-i\frac{p^+}{k^+}k\cdot(y'-z)}$$

Net factor $1 \times \frac{k^+}{S} = \frac{k^+}{S}$; the prefactor $\frac{1}{k^+} \frac{p^+}{S}$ becomes $\frac{p^+}{S^2}$ — the $\frac{1}{k^+}$ out front cancels, exactly as in your second equation. Pushing $\frac{p^+}{S^2}$ into the bracket:

group weight before	$\times p^+ / S^2$	you wrote	diff
$d_\perp$	$d_\perp p^+ / S^2$	same	0
S / k^+	$p^+ / (k^+ S)$	same	0
S / p^+	$1 / S$	same	0
$-S^2 / p^{+2}$	$-1 / p^+$	same	0
$-S^2 / k^{+2}$	$-p^+ / k^{+2}$	same	0
$-S^2 / (p^+ k^+)$	$-1 / k^+$	same	0

5. The measure

Both displayed equations read $\int_{w,w',y',z,y,x}$. Three problems: **y never appears** anywhere in the integrand (there is no y in this row — only y'); **x' is missing**, although it appears in $\frac{(x'-y')^k}{(x'-y')^2}$ and in $\rho^{e'}(x')$; and in the *second* equation w' has already been integrated out. It should be $\int_{w,w',z,x,x',y'}$ before and $\int_{w,z,x,x',y'}$ after.

6. Colour

Your bra writes $U^{e'c'}(y')$ and $U^{db}(z)$ in term 1 but $U^{c'e'}(x')$ and $U^{bd'}(z)$ in term 2 — both transposed between the two terms. Fix them to a single convention (f -index first, say). Then:

product	collapses?
$\text{bra}_1 \times \text{ket}_1$	no
$\text{bra}_1 \times \text{ket}_2$	no
$\text{bra}_2 \times \text{ket}_1$	no
$\text{bra}_2 \times \text{ket}_2$	yes $\rightarrow -N_c [U^\dagger(x')U(x)]^{e'e}$ (residual $5 \cdot 10^{-15}$)

Unlike the two- $\mathcal{C}$ row, there is no overall N_c here. The collapse needs $\sum_b U^b(z)U^b(z) = \delta$, i.e. a Wilson line at the *same* point on the *same* free index on both sides. On the bra side a always sits on f and b on $U(z)$. On the ket side, term 1 has a on $U(w)$ and no $U(z)$ at all; only term 2 matches the bra's pattern. So three of the four products keep their full Wilson-line content and must be carried along individually.

7. Transverse divergences: the short-distance scan

Scanning the corrected, w' -integrated integrand over $\{x, x', y', z, w\}$, all 26 collapsing subsets:

no short-distance divergence — no UV.
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The four candidate regions behave as they must: $y' \rightarrow z$ and $y' \rightarrow x'$ are ρ^{-1} (one kernel each); $z \rightarrow w$ is ρ^{-1} (the $\frac{1}{\mathcal{C}}$ numerators vanish); $x \rightarrow w$ and $x \rightarrow z$ are ρ^{-1} . A logarithm would need ρ^{-2} , i.e. two kernels singular on the same separation, and there are none.

8. The large–distance ends — and the key finding

variable $\rightarrow \infty$	with the phases	phases switched off	reading
x	0.16433, 0.16437, 0.16437 — log	same	x has no phase; cut by the target (it is the argument of ρ)
x'	$\sim 10^{-13}$	$\sim 10^{-13}$	convergent
y'	$7.2 \cdot 10^{-3} \rightarrow 6.9 \cdot 10^{-4}$	0.23246 flat	log, Fresnel–damped by $e^{-i\frac{k}{k^+}y'}$
z	decreasing / noisy	0.010124 flat	log, damped by $e^{+i\frac{p^+}{k^+}k \cdot z}$
w	oscillatory	0.37424 flat	log, damped by $e^{+ik \cdot w}$

8.1 Which part of the bracket makes the z logarithm

Measuring $\langle \text{group} \rangle \times |z|^2$ group by group (phase off):

group	$ z = 10^2$	10^3	10^4	behaviour
$d_\perp$	$8.17 \cdot 10^{-6}$	$8.19 \cdot 10^{-8}$	$8.19 \cdot 10^{-10}$	falls faster
S/k^+	$1.2795 \cdot 10^{-3}$	$1.3018 \cdot 10^{-3}$	$1.3021 \cdot 10^{-3}$	log
S/p^+	$-1.42 \cdot 10^{-5}$	$-1.42 \cdot 10^{-7}$	$-1.42 \cdot 10^{-9}$	falls faster
$-1/p^+$	$-6.92 \cdot 10^{-6}$	$-6.92 \cdot 10^{-8}$	$-6.92 \cdot 10^{-10}$	falls faster
$-p^+/k^{+2}$	$-1.761 \cdot 10^{-3}$	$-1.784 \cdot 10^{-3}$	$-1.784 \cdot 10^{-3}$	log
$-1/k^+$	$-9.560 \cdot 10^{-3}$	$-9.642 \cdot 10^{-3}$	$-9.642 \cdot 10^{-3}$	log

The mechanism is that $(x - z)^{k'}$ and $(x - z)^m$ grow like $|z|$ while $N^m = \frac{(y' - z)^m}{(y' - z)^2}$ falls like $1/|z|$ and $\frac{1}{D} \rightarrow \frac{1}{p^+|z|^2}$; the net is $|z|^{-2}$.

8.2 The coefficient of that logarithm carries $1/p^+$

Because $D = p^+A + k^+B \rightarrow p^+|z|^2$ at large $|z|$:

p^+	0.02	0.05	0.2	1.0	3.0
coefficient	−0.17842	−0.07144	−0.01814	−0.00446	−0.00253
$\times p^+$	−0.003568	−0.003572	−0.003627	−0.004460	−0.007581

Flat at small p^+ : the coefficient goes like $1/p^+$.

8.3 The phase sets the upper end — and it moves with p^+

The z phase is $e^{+i\frac{p^+}{k^+}k \cdot z}$, so the log is cut at $|z| \sim \frac{k^+}{p^+|k_\perp|}$. Integrating $\int_{|z| < R} d^2z$ with the phase on:

p^+	$k^+/(p^+ k_\perp)$	$R = 10^2$	10^3	10^4
0.20	7.2	1.7966	1.7946	1.8091
1.00	1.4	1.6860	1.6748	1.6858
4.00	0.4	0.5021	0.4886	0.4850

Saturated in every case — **there is no collinear pole to regulate**. The logarithmic window is

$$\sqrt{\frac{k^+(x - w)^2}{p^+}} < |z| < \frac{k^+}{p^+|k_\perp|} \implies \text{extent} \simeq \frac{1}{2} \log \frac{k^+}{p^+} + \text{const}$$

and the measured extents at $p^+ = 10^{-1}, 10^{-2}, 10^{-3}$ are 1.027, 2.179, 3.330 — increments of $1.151 = \frac{1}{2} \log 10$ exactly, confirming the $\frac{1}{2} \log \frac{k^+}{p^+}$ law.

9. Consequence: a double logarithm

Integrating z completely (phase on, $R = 60 k^+/(p^+|k_\perp|)$) and fitting:

resolution	$I(10^{-3})$	fit of $I(p^+) p^+$
$R_{\text{fac}} = 40, n_r = 1200$	-200.73	$0.11737 - 0.05064 \log \frac{k^+}{p^+}$
$R_{\text{fac}} = 80, n_r = 1800$	-200.17	$0.11120 - 0.04950 \log \frac{k^+}{p^+}$
$R_{\text{fac}} = 160, n_r = 2400$	-200.22	$0.10851 - 0.04900 \log \frac{k^+}{p^+}$

The log coefficient is stable to 3% across a factor of four in cutoff and a doubling of the grid, so it is not a numerical artefact. Hence

$$I(p^+) \simeq \frac{1}{p^+} \left[a + b \log \frac{k^+}{p^+} \right] \implies \int_{\Lambda} \frac{dp^+}{p^+} \left[a + b \log \frac{k^+}{p^+} \right] = a \log \frac{k^+}{\Lambda} + \frac{b}{2} \log^2 \frac{k^+}{\Lambda}$$

a **soft-collinear double logarithm**. It arises because the two regions are not independent here: $\frac{1}{D}$ supplies the $\frac{1}{p^+}$ and the phase's cutoff, which supplies the collinear log, itself grows like $1/p^+$.

10. Your three questions, answered

10.1 Is there a UV divergence to put in $\overline{\text{MS}}$?

No. All 26 short-distance regions are convergent (§7). There is no $\frac{1}{\epsilon}$ in this row, so dimensional regularisation has nothing to do and $\overline{\text{MS}}$ has nothing to subtract.

10.2 Is there a collinear divergence to absorb into a fragmentation function?

No — not at fixed p^+ . There is a collinear region (large $|z|$), and it is the same region that gave the fragmentation pole in the two- $\mathfrak{C}$ row. But there the z integral was **phase-free** (both $\mathfrak{C}$'s deltas had been used, leaving the phase in $y' - y$ only), so it produced a genuine $\int_{z^2}^{\infty} \frac{d^2 z}{z^2}$ and a $\frac{1}{\epsilon}$. Here only *one* $\mathfrak{C}$ delta exists, the other side is $\mathfrak{B}_2$, and the leftover phase $e^{+i\frac{p^+}{k^+} k \cdot z}$ sits directly on z and cuts the integral off. The result is a finite logarithm of a physical ratio, not a pole. **Nothing to subtract into $D_{g/g}$ from this row.**

10.3 Can you use the same + prescription for the p^+ integration?

No. The + prescription is defined by

$$\int_0^P dp^+ \left[\frac{1}{p^+} \right]_+ g(p^+) = \int_0^P dp^+ \frac{g(p^+) - g(0)}{p^+}$$

and it is meaningful only if $g(0)$ **exists**. Here $g(p^+) = p^+ I(p^+)$ and §9 shows $g(p^+) = a + b \log \frac{k^+}{p^+}$, which **diverges logarithmically** as $p^+ \rightarrow 0$. There is no $g(0)$ to subtract, so the naive prescription is ill-defined: it would leave you subtracting infinity from infinity.

Contrast the two- $\mathfrak{C}$ row, where I told you the + prescription is legitimate: there, after the z integral, the only ζ -dependence left in g was the smooth phase $e^{-ik \cdot r / \zeta}$, analytic at the endpoint. The difference is not the prescription — it is whether anything singular survives in g at the subtraction point.

What you can do instead. Two routes, both standard:

- Two-step subtraction.** Split $g(p^+) = b \log \frac{k^+}{p^+} + \tilde{g}(p^+)$ with $\tilde{g}$ now regular at the endpoint. The second piece takes an ordinary $\left[\frac{1}{p^+} \right]_+$; the first integrates to the explicit $\frac{b}{2} \log^2 \frac{k^+}{\Lambda}$, which you keep as a $\log^2$ and match against the partner rows.
- Keep Λ .** Do not prescribe anything; carry $\log \frac{k^+}{\Lambda}$ and $\log^2 \frac{k^+}{\Lambda}$ explicitly. A $\log^2$ at this order is the signature of an $\mathcal{O}(\alpha_s)$ correction to the evolution kernel, so it has to be matched, not prescribed away.

One honest caveat. The $\log^2$ is established for *this* interference. It is one of a conjugate pair (its partner $\mathfrak{B}_2^\dagger \times \mathfrak{A}\mathfrak{C}$ is the complex conjugate, so the pair gives 2Re and the double log doubles rather than cancels). Whether it survives in the full NLO sum depends on the remaining rows — that cannot be decided from this row alone, and I have not checked it.

11. Checklist of things to fix

1. The $-\frac{S^2}{p+2}$ group: $(z-w)^{k'} \rightarrow (x-w)^{k'}$ in the second term (§3).
2. Add a line stating the delta rescaling, or the prefactor will look wrong by S^2/k^{+2} (§1).
3. Measure: drop $\int_y$, add $\int_{x'}$, drop $\int_{w'}$ from the second equation (§5).
4. Make the Wilson–line index orders uniform in the bra (§6).
5. $d_\perp = 2$ is safe here, since there is no $\frac{1}{\epsilon}$ for its $O(\epsilon)$ part to hit.

Checked by `src/row25_check.py`, `row25_col2.py`, `row25_div.py`, `row25_quant2.py`, `row25_soft.py`, `row25_conv.py`; output in `RESULTS_row25.txt`.